Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030078

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Query 1: -CREATE DATABASE

```
CREATE DATABASE BankDB1;
USE BankDB1;
```

Output: -

✓	45	16:49:53	CREATE DATABASE BankDB1	1 row(s) affected	0.016 sec
✓	46	16:49:57	USE BankDB1	0 row(s) affected	0.000 sec

Query 2: -CREATE CUSTOMER TABLE

```
CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY,
    Customer_Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(15),
    Email VARCHAR(100),
    City VARCHAR(50)
);
```

Output: -

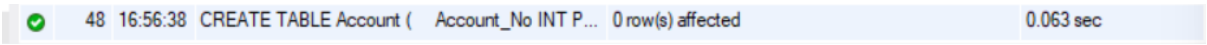
✓	47	16:51:31	CREATE TABLE Customer (	Customer_ID INT ...	0 row(s) affected

Query 3: -CREATE ACCOUNT TABLE

```sql
CREATE TABLE Account (
    Account_No INT PRIMARY KEY,
    Customer_ID INT,
    Account_Type VARCHAR(20),
    Balance DECIMAL(12,2),
    Branch VARCHAR(50),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
```

Output: -

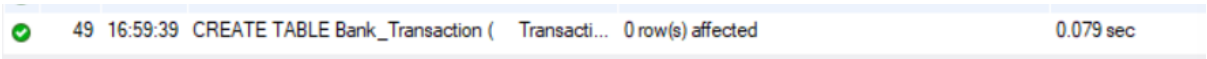


Query 4: -CREATE BANK TRANSACTION TABLE

```sql
CREATE TABLE Bank_Transaction (
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (Account_No)
    REFERENCES Account(Account_No)
);
```

Output: -



Query 5: - CREATE LOAN TABLE

```sql
CREATE TABLE Loan (
    Loan_ID INT PRIMARY KEY,
    Customer_ID INT,
    Loan_Type VARCHAR(30),
    Loan_Amount DECIMAL(12,2),
    Interest_Rate DECIMAL(5,2),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
```

Output: -

✔	50	17:01:41	CREATE TABLE Loan (Loan_ID INT PRIMAR...	0 row(s) affected	0.109 sec

Query 6: -INSERT CUSTOMER DATA

```sql
INSERT INTO Customer
(Customer_ID, Customer_Name, Phone, Email, City)
VALUES
(101, 'Aswitha', '9876543210',
 'aswitha@gmail.com', 'Hyderabad'),
(102, 'Priyadarshini', '9876543211',
 'priya@gmail.com', 'Vijayawada'),
(103, 'Adhritha', '9876543212',
 'adhritha@gmail.com', 'Bangalore'),
(104, 'Akshara', '9876543213',
 'akshara@gmail.com', 'Chennai'),
(105, 'Sambaiah', '9876543214',
 'sam@gmail.com', 'Hyderabad'),
(106, 'Ananya Varshitha]', '9876543215',
 'ananya@gmail.com', 'Delhi'),
(107, 'Adithya Vardan', '9876543216',
 'adithya@gmail.com', 'Mumbai'),
(108, 'Yashwanth', '9876543217',
 'yashwanth@gmail.com', 'Pune'),
(109, 'Nihita', '9876543218',
 'nihita@gmail.com', 'Hyderabad'),
(110, 'Rushika', '9876543219',
 'rushika@gmail.com', 'Vijayawada');
```

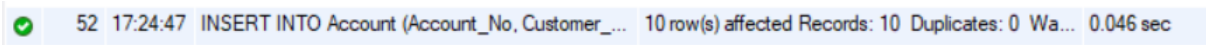
Output: -

✔	51	17:20:23	INSERT INTO Customer (Customer_ID, Customer...	10 row(s) affected Records: 10 Duplicates: 0 Wa...	0.031 sec

Query 7: -INSERT ACCOUNT DATA

```
INSERT INTO Account
(Account_No, Customer_ID, Account_Type, Balance, Branch)
VALUES
(10001, 101, 'Savings', 50000, 'Hyderabad'),
(10002, 102, 'Savings', 75000, 'Vijayawada'),
(10003, 103, 'Current', 120000, 'Bangalore'),
(10004, 104, 'Savings', 45000, 'Chennai'),
(10005, 105, 'Current', 90000, 'Hyderabad'),
(10006, 106, 'Savings', 65000, 'Delhi'),
(10007, 107, 'Current', 150000, 'Mumbai'),
(10008, 108, 'Savings', 35000, 'Pune'),
(10009, 109, 'Savings', 85000, 'Hyderabad'),
(10010, 110, 'Current', 110000, 'Vijayawada');
```

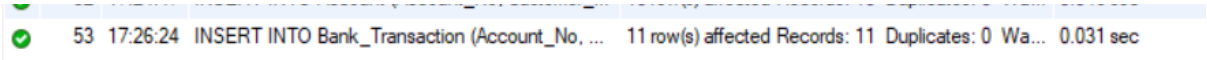
Output: -



Query 8: -INSERT TRANSACTION DATA

```
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),
(10001, 'WITHDRAW', 5000),
(10002, 'DEPOSIT', 15000),
(10003, 'WITHDRAW', 20000),
(10004, 'DEPOSIT', 5000),
(10005, 'WITHDRAW', 10000),
(10006, 'DEPOSIT', 12000),
(10007, 'DEPOSIT', 25000),
(10008, 'WITHDRAW', 5000),
(10009, 'DEPOSIT', 20000),
(10010, 'WITHDRAW', 15000);
```

Output: -



Query 9: -INSERT LOAN DATA

```
INSERT INTO Loan
(Loan_ID, Customer_ID, Loan_Type,
 Loan_Amount, Interest_Rate)
VALUES
(501, 101, 'Home Loan', 5000000, 7.5),

(502, 102, 'Education Loan', 1000000, 6.5),

(503, 103, 'Car Loan', 800000, 8.2),

(504, 104, 'Personal Loan', 500000, 10.5),

(505, 105, 'Home Loan', 4000000, 7.2),

(506, 106, 'Car Loan', 900000, 8.5),

(507, 107, 'Business Loan', 3000000, 9.0),

(508, 109, 'Personal Loan', 600000, 10.0);
```

Output: -

✔	54	17:27:34	INSERT INTO Loan (Loan_ID, Customer_ID, Loa...	8 row(s) affected Records: 8	Duplicates: 0	Warni...	0.047 sec

Query 10: - DISPLAY ORIGINAL TABLES

```
SELECT * FROM Customer;
```

Customer_ID	Customer_Name	Phone	Email	City
101	Aswitha	9876543210	aswitha@gmail.com	Hyderabad
102	Priyadarshini	9876543211	priya@gmail.com	Vijayawada
103	Adhritha	9876543212	adhritha@gmail.com	Bangalore
104	Akshara	9876543213	akshara@gmail.com	Chennai
105	Sambaiah	9876543214	sam@gmail.com	Hyderabad
106	Ananya Varshitha]	9876543215	ananya@gmail.com	Delhi
107	Adithya Vardan	9876543216	adithya@gmail.com	Mumbai
108	Yashwanth	9876543217	yashwanth@gmail.com	Pune
109	Nihita	9876543218	nihita@gmail.com	Hyderabad
110	Rushika	9876543219	rushika@gmail.com	Vijayawada
NULL	NULL	NULL	NULL	NULL

```
SELECT * FROM Account;
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
10006	106	Savings	65000.00	Delhi
10007	107	Current	150000.00	Mumbai
10008	108	Savings	35000.00	Pune
10009	109	Savings	85000.00	Hyderabad
10010	110	Current	110000.00	Vijayawada
NULL	NULL	NULL	NULL	NULL

```
SELECT * FROM Bank_Transaction;
SELECT * FROM Loan;
```

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-15 17:26:24
2	10001	WITHDRAW	5000.00	2026-08-15 17:26:24
3	10002	DEPOSIT	15000.00	2026-08-15 17:26:24
4	10003	WITHDRAW	20000.00	2026-08-15 17:26:24
5	10004	DEPOSIT	5000.00	2026-08-15 17:26:24
6	10005	WITHDRAW	10000.00	2026-08-15 17:26:24
7	10006	DEPOSIT	12000.00	2026-08-15 17:26:24
8	10007	DEPOSIT	25000.00	2026-08-15 17:26:24
9	10008	WITHDRAW	5000.00	2026-08-15 17:26:24
10	10009	DEPOSIT	20000.00	2026-08-15 17:26:24
11	10010	WITHDRAW	15000.00	2026-08-15 17:26:24
NULL	NULL	NULL	NULL	NULL

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50
505	105	Home Loan	4000000.00	7.20
506	106	Car Loan	900000.00	8.50
507	107	Business Loan	3000000.00	9.00
508	109	Personal Loan	600000.00	10.00
NULL	NULL	NULL	NULL	NULL

Query 11: -CREATE A VIEW FOR ALL CUSTOMERS

```
CREATE VIEW Customer_View AS
SELECT *
FROM Customer;
SELECT * FROM Customer_View;
```

Output: -

Customer_ID	Customer_Name	Phone	Email	City
101	Aswitha	9876543210	aswitha@gmail.com	Hyderabad
102	Priyadarshini	9876543211	priya@gmail.com	Vijayawada
103	Adhritha	9876543212	adhritha@gmail.com	Bangalore
104	Akshara	9876543213	akshara@gmail.com	Chennai
105	Sambaiah	9876543214	sam@gmail.com	Hyderabad
106	Ananya Varshitha]	9876543215	ananya@gmail.com	Delhi
107	Adithya Vardan	9876543216	adithya@gmail.com	Mumbai
108	Yashwanth	9876543217	yashwanth@gmail.com	Pune
109	Nihita	9876543218	nihita@gmail.com	Hyderabad
110	Rushika	9876543219	rushika@gmail.com	Vijayawada

Customer_View 5 ×

Output

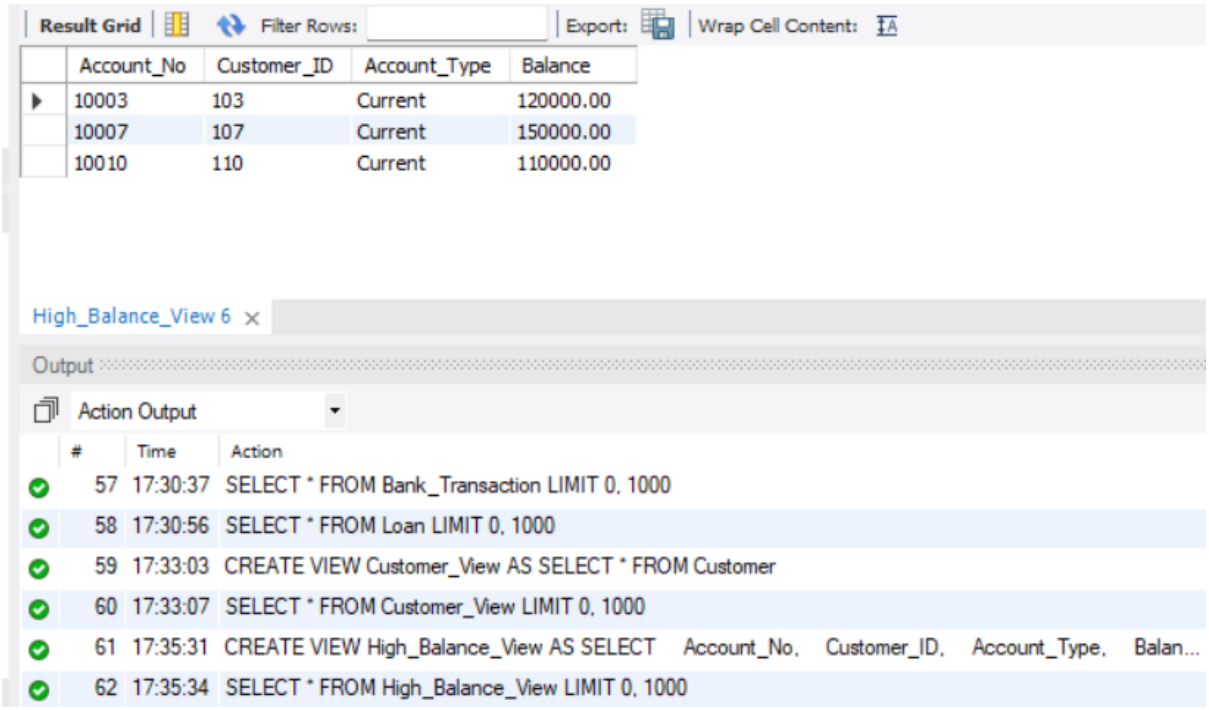
Action Output

#	Time	Action
55	17:28:42	SELECT * FROM Customer LIMIT 0, 1000
56	17:29:56	SELECT * FROM Account LIMIT 0, 1000
57	17:30:37	SELECT * FROM Bank_Transaction LIMIT 0, 1000
58	17:30:56	SELECT * FROM Loan LIMIT 0, 1000
59	17:33:03	CREATE VIEW Customer_View AS SELECT * FROM Customer
60	17:33:07	SELECT * FROM Customer_View LIMIT 0, 1000

Query 12: -HIGH BALANCE ACCOUNT VIEW

```
CREATE VIEW High_Balance_View AS
SELECT
    Account_No,
    Customer_ID,
    Account_Type,
    Balance
FROM Account
WHERE Balance > 100000;
SELECT * FROM High_Balance_View;
```

Output: -



Query 13: -VIEWS USING JOINS

CUSTOMER + ACCOUNT VIEW

```
CREATE VIEW Customer_Account_View AS
SELECT
    C.Customer_ID,
    C.Customer_Name,
    C.City,
    A.Account_No,
    A.Account_Type,
    A.Balance,
    A.Branch
FROM Customer C
JOIN Account A
ON C.Customer_ID = A.Customer_ID;
SELECT * FROM Customer_Account_View;
```

Output: -

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Branch
101	Aswitha	Hyderabad	10001	Savings	50000.00	Hyderabad
102	Priyadarshini	Vijayawada	10002	Savings	75000.00	Vijayawada
103	Adhritha	Bangalore	10003	Current	120000.00	Bangalore
104	Akshara	Chennai	10004	Savings	45000.00	Chennai
105	Sambaiah	Hyderabad	10005	Current	90000.00	Hyderabad
106	Ananya Varshitha]	Delhi	10006	Savings	65000.00	Delhi
107	Adithya Vardan	Mumbai	10007	Current	150000.00	Mumbai
108	Yashwanth	Pune	10008	Savings	35000.00	Pune
109	Nihita	Hyderabad	10009	Savings	85000.00	Hyderabad
110	Rushika	Vijayawada	10010	Current	110000.00	Vijayawada

Customer_Account_View 7 ×

Output

Action Output

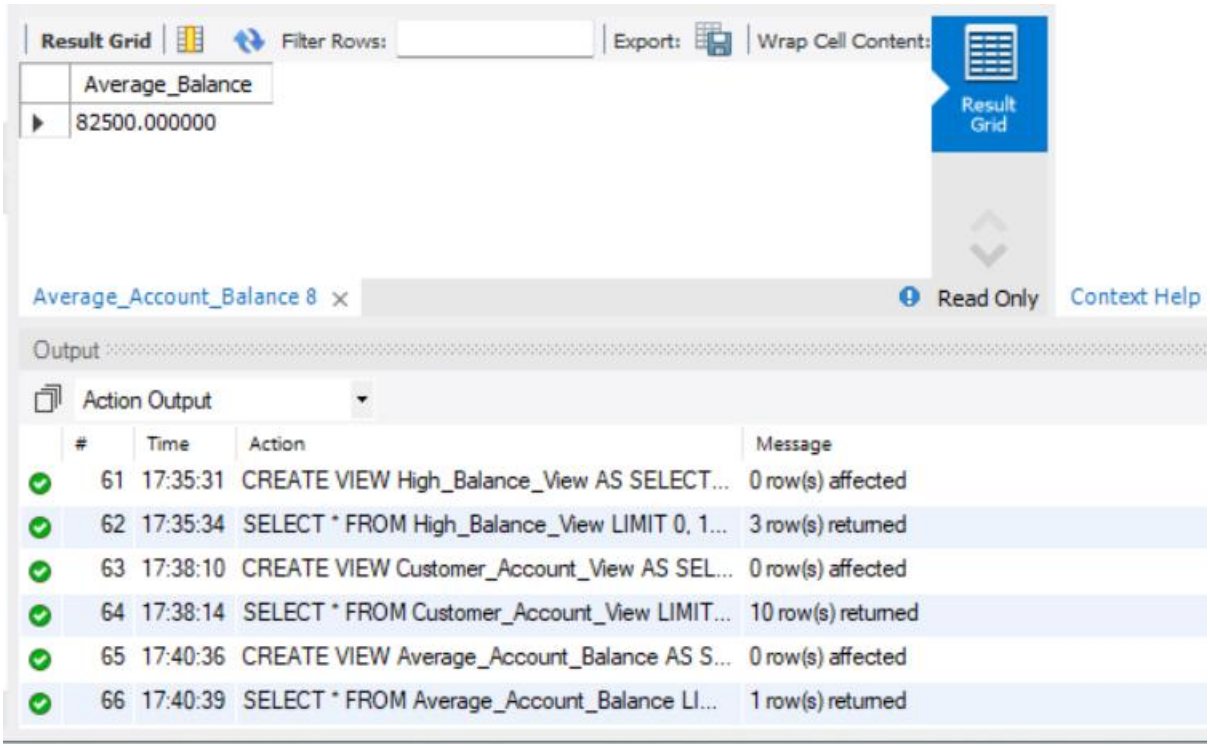
#	Time	Action	Message
59	17:33:03	CREATE VIEW Customer_View AS SELECT * FROM Customer	0 row(s) affected
60	17:33:07	SELECT * FROM Customer_View LIMIT 0, 1000	10 row(s) returned
61	17:35:31	CREATE VIEW High_Balance_View AS SELECT Account_No, Customer_ID, Account_Type, Balan...	0 row(s) affected
62	17:35:34	SELECT * FROM High_Balance_View LIMIT 0, 1000	3 row(s) returned
63	17:38:10	CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, A....	0 row(s) affected
64	17:38:14	SELECT * FROM Customer_Account_View LIMIT 0, 1000	10 row(s) returned

Query 14: -VIEWS USING AGGREGATE FUNCTIONS

AVERAGE ACCOUNT BALANCE

```
CREATE VIEW Average_Account_Balance AS
SELECT
    AVG(Balance) AS Average_Balance
FROM Account;
SELECT * FROM Average_Account_Balance;
```

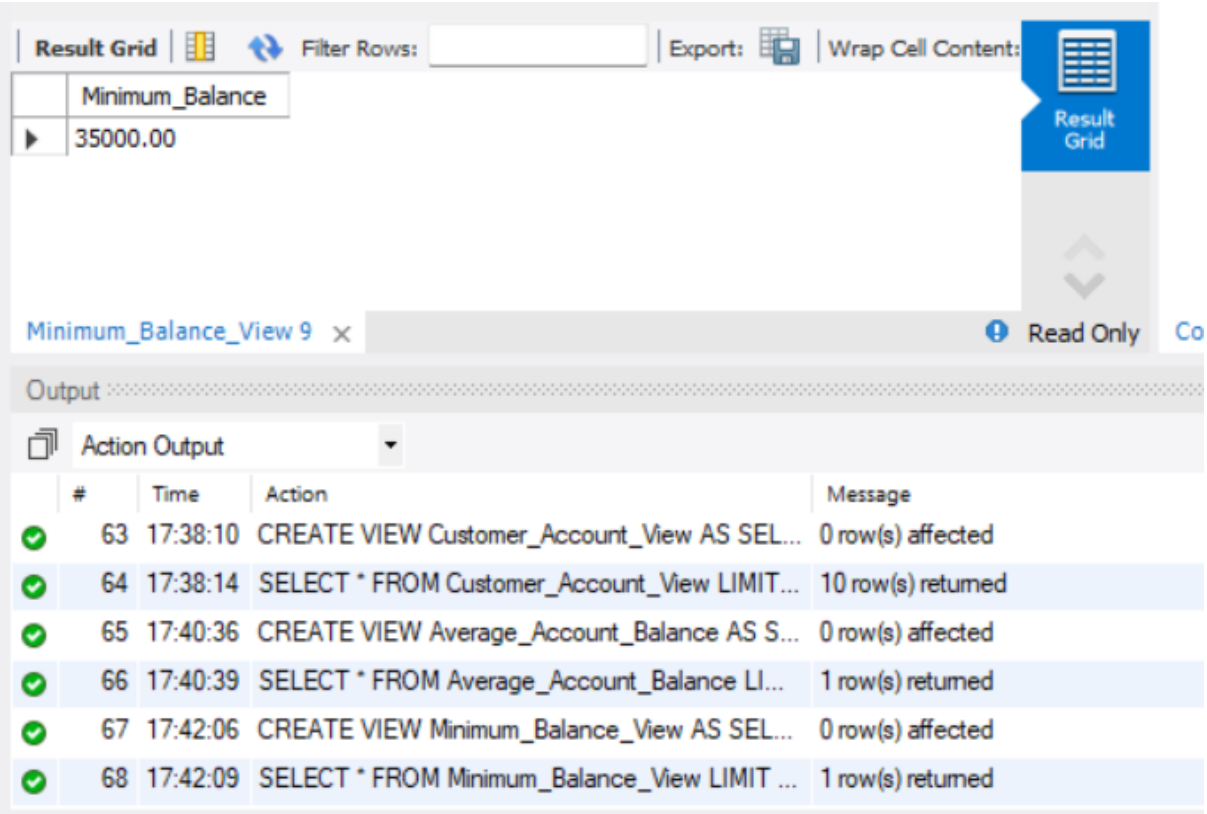
Output: -



Query 15: -MINIMUM ACCOUNT BALANCE

```
CREATE VIEW Minimum_Balance_View AS
SELECT
    MIN(Balance) AS Minimum_Balance
FROM Account;
SELECT * FROM Minimum_Balance_View;
```

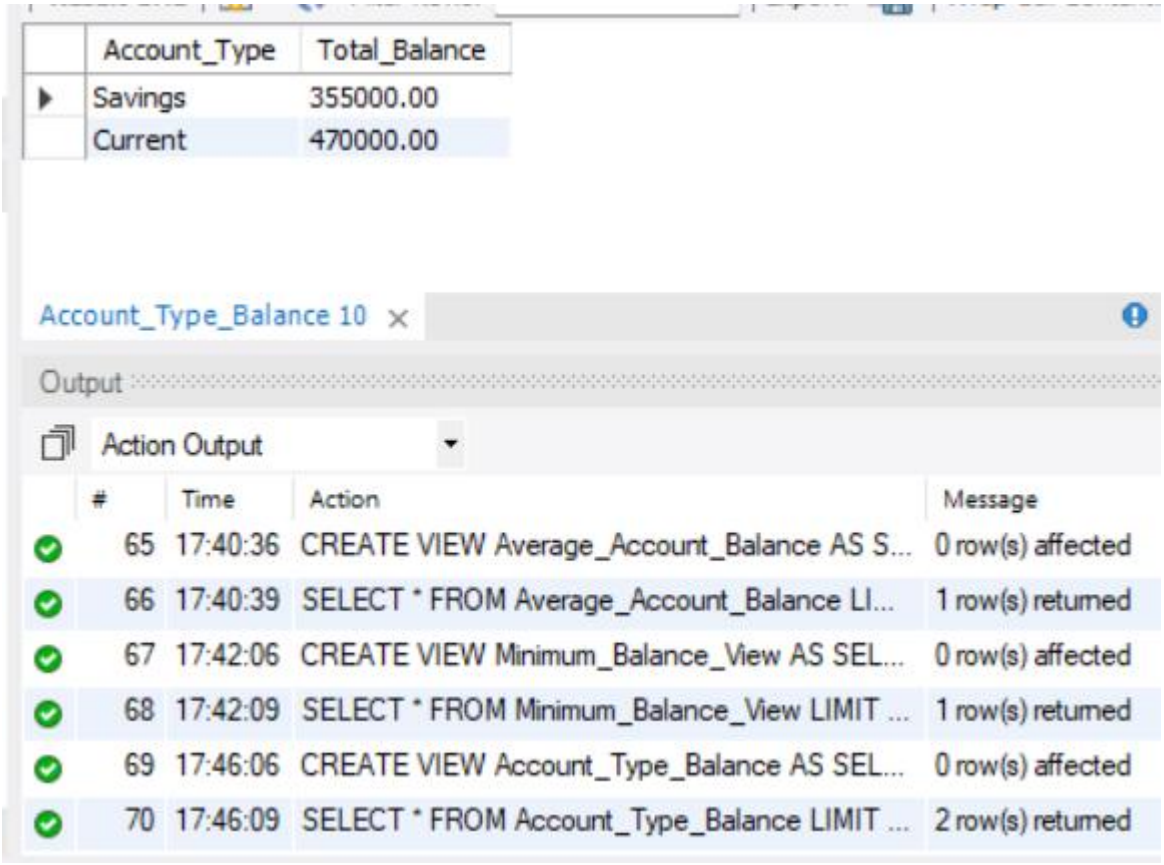
Output: -



Query 16: -ACCOUNT TYPE TOTAL BALANCE

```
CREATE VIEW Account_Type_Balance AS
SELECT
    Account_Type,
    SUM(Balance) AS Total_Balance
FROM Account
GROUP BY Account_Type;
SELECT * FROM Account_Type_Balance;
```

Output: -

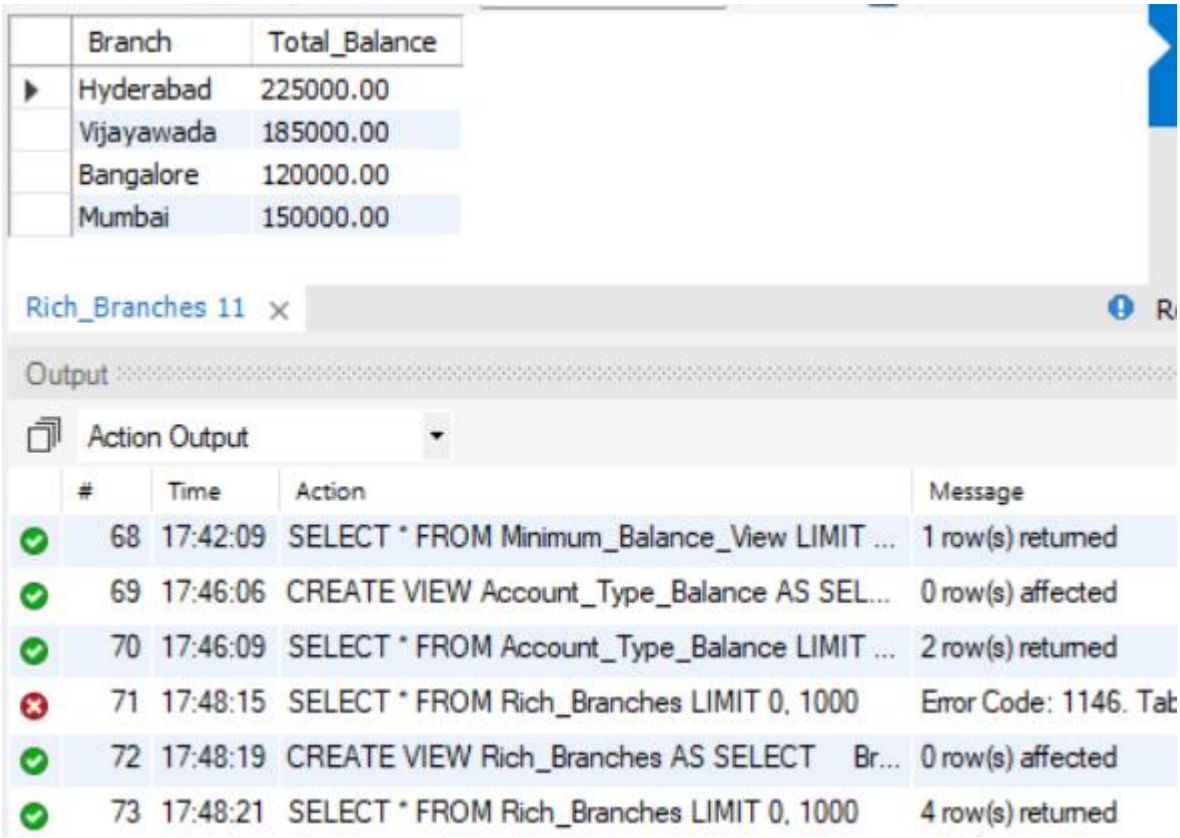


Query 17: -HAVING IN VIEWS

BRANCHES WITH TOTAL BALANCE ABOVE 100000

```
CREATE VIEW Rich_Branches AS
SELECT
    Branch,
    SUM(Balance) AS Total_Balance
FROM Account
GROUP BY Branch
HAVING SUM(Balance) > 100000;
SELECT * FROM Rich_Branches;
```

Output: -



Branch	Total_Balance
Hyderabad	225000.00
Vijayawada	185000.00
Bangalore	120000.00
Mumbai	150000.00

Rich_Branches 11 ×

Output

Action Output

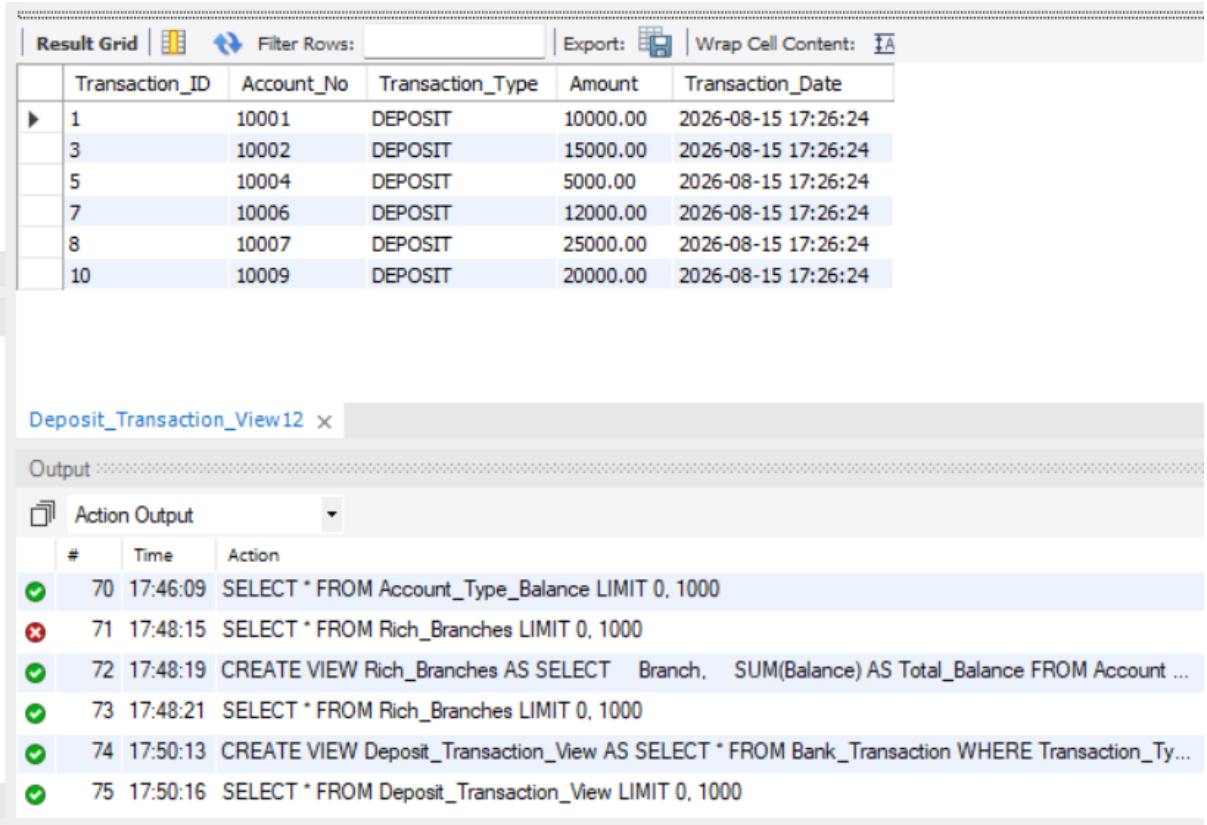
#	Time	Action	Message
68	17:42:09	SELECT * FROM Minimum_Balance_View LIMIT ...	1 row(s) returned
69	17:46:06	CREATE VIEW Account_Type_Balance AS SEL...	0 row(s) affected
70	17:46:09	SELECT * FROM Account_Type_Balance LIMIT ...	2 row(s) returned
71	17:48:15	SELECT * FROM Rich_Branches LIMIT 0, 1000	Error Code: 1146. Tab
72	17:48:19	CREATE VIEW Rich_Branches AS SELECT Br...	0 row(s) affected
73	17:48:21	SELECT * FROM Rich_Branches LIMIT 0, 1000	4 row(s) returned

Query 18: -TRANSACTION VIEWS

DEPOSIT TRANSACTION VIEW

```
CREATE VIEW Deposit_Transaction_View AS
SELECT *
FROM Bank_Transaction
WHERE Transaction_Type = 'DEPOSIT';
SELECT * FROM Deposit_Transaction_View;
```

Output: -



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

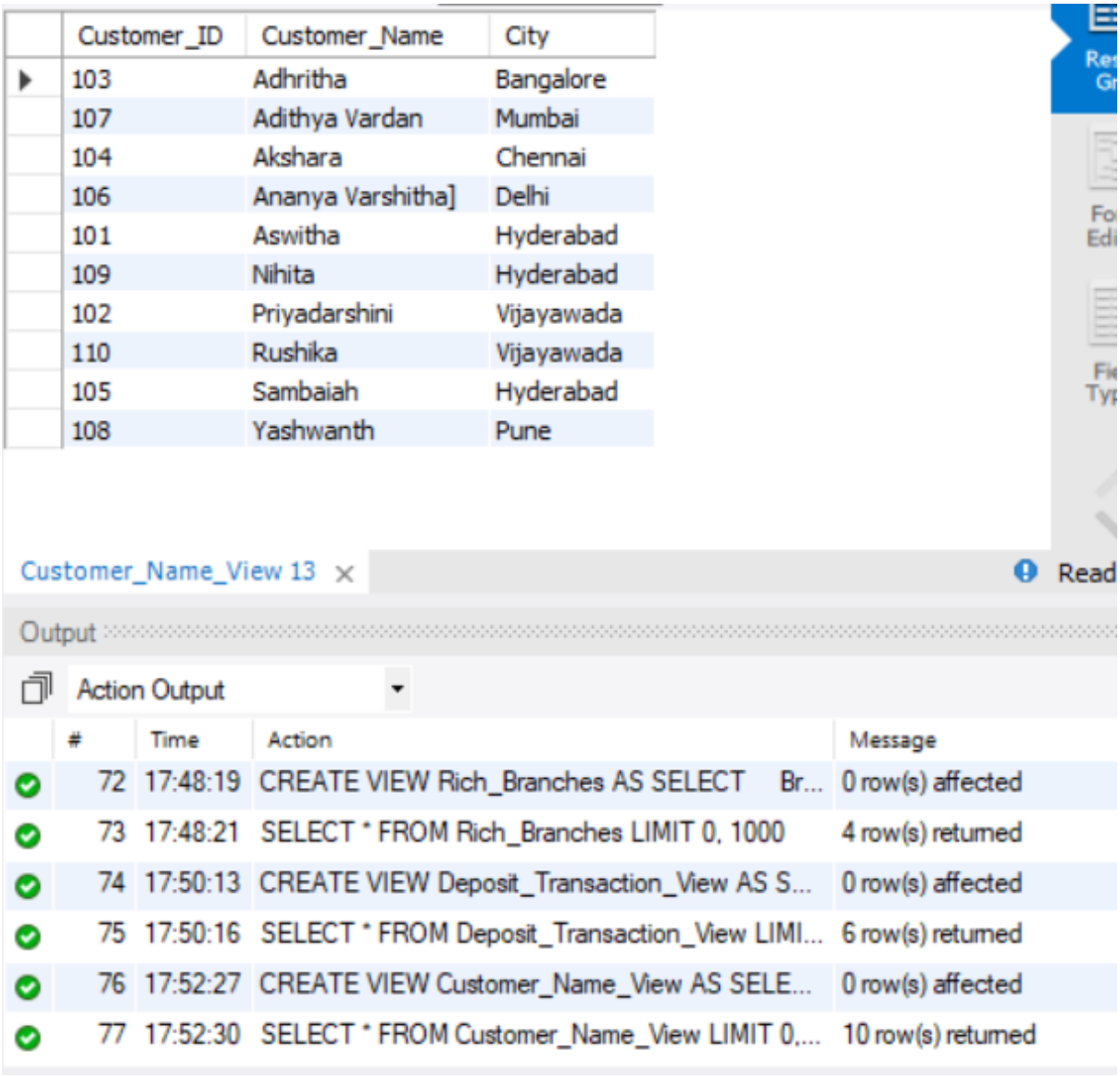
Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-15 17:26:24
3	10002	DEPOSIT	15000.00	2026-08-15 17:26:24
5	10004	DEPOSIT	5000.00	2026-08-15 17:26:24
7	10006	DEPOSIT	12000.00	2026-08-15 17:26:24
8	10007	DEPOSIT	25000.00	2026-08-15 17:26:24
10	10009	DEPOSIT	20000.00	2026-08-15 17:26:24

Query 19: -VIEWS WITH ORDER BY

CUSTOMERS ORDERED BY NAME

```
CREATE VIEW Customer_Name_View AS
SELECT
    Customer_ID,
    Customer_Name,
    City
FROM Customer
ORDER BY Customer_Name;
SELECT * FROM Customer_Name_View;
```

Output: -



Customer_ID	Customer_Name	City
103	Adhritha	Bangalore
107	Adithya Vardan	Mumbai
104	Akshara	Chennai
106	Ananya Varshitha]	Delhi
101	Aswitha	Hyderabad
109	Nihita	Hyderabad
102	Priyadarshini	Vijayawada
110	Rushika	Vijayawada
105	Sambaiah	Hyderabad
108	Yashwanth	Pune

Query 20: -VIEW WITH CALCULATED COLUMN

LOAN TOTAL AMOUNT VIEW

```sql
CREATE VIEW Loan_Total_Amount_View AS
SELECT
    Loan_ID,
    Customer_ID,
    Loan_Type,
    Loan_Amount,
    Interest_Rate,
    Loan_Amount +
    (Loan_Amount * Interest_Rate / 100)
    AS Total_Amount
FROM Loan;
SELECT * FROM Loan_Total_Amount_View;
```

Output: -

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate	Total_Amount
501	101	Home Loan	5000000.00	7.50	5375000.00000000
502	102	Education Loan	1000000.00	6.50	1065000.00000000
503	103	Car Loan	800000.00	8.20	865600.00000000
504	104	Personal Loan	500000.00	10.50	552500.00000000
505	105	Home Loan	4000000.00	7.20	4288000.00000000
506	106	Car Loan	900000.00	8.50	976500.00000000
507	107	Business Loan	3000000.00	9.00	3270000.00000000
508	109	Personal Loan	600000.00	10.00	660000.00000000

Loan_Total_Amount_View 14 ×

Output

Action Output

#	Time	Action
74	17:50:13	CREATE VIEW Deposit_Transaction_View AS SELECT * FROM Bank_Transaction WHERE Transactio
75	17:50:16	SELECT * FROM Deposit_Transaction_View LIMIT 0, 1000
76	17:52:27	CREATE VIEW Customer_Name_View AS SELECT Customer_ID, Customer_Name, City FROM C
77	17:52:30	SELECT * FROM Customer_Name_View LIMIT 0, 1000
78	17:55:12	CREATE VIEW Loan_Total_Amount_View AS SELECT Loan_ID, Customer_ID, Loan_Type, L
79	17:55:15	SELECT * FROM Loan_Total_Amount_View LIMIT 0, 1000

Query 21: -USING VIEW WITH QUERIES

DISPLAY CUSTOMER ACCOUNTS ORDERED BY BALANCE

```sql
SELECT *
FROM Customer_Account_View
ORDER BY Balance DESC;
```

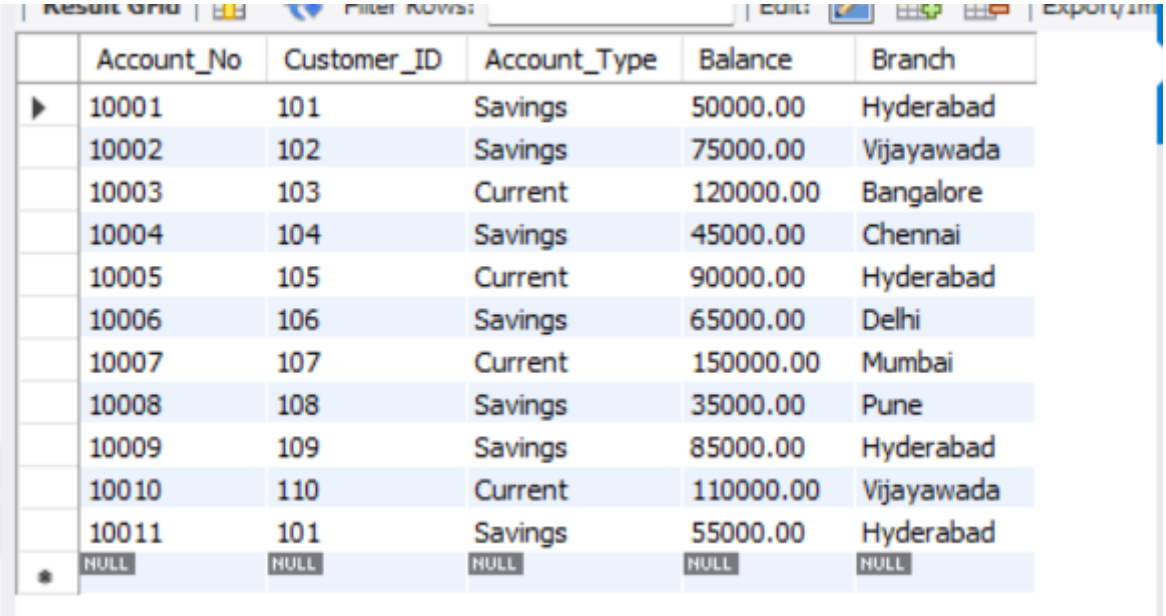
Output: -

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Branch
107	Adithya Vardan	Mumbai	10007	Current	150000.00	Mumbai
103	Adhritha	Bangalore	10003	Current	120000.00	Bangalore
110	Rushika	Vijayawada	10010	Current	110000.00	Vijayawada
105	Sambaiah	Hyderabad	10005	Current	90000.00	Hyderabad
109	Nihita	Hyderabad	10009	Savings	85000.00	Hyderabad
102	Priyadarshini	Vijayawada	10002	Savings	75000.00	Vijayawada
106	Ananya Varshitha]	Delhi	10006	Savings	65000.00	Delhi
101	Aswitha	Hyderabad	10001	Savings	50000.00	Hyderabad
104	Akshara	Chennai	10004	Savings	45000.00	Chennai
108	Yashwanth	Pune	10008	Savings	35000.00	Pune

Query 22: -INSERT THROUGH VIEW

```
CREATE VIEW Simple_Account_View AS
SELECT
    Account_No,
    Customer_ID,
    Account_Type,
    Balance,
    Branch
FROM Account;
INSERT INTO Simple_Account_View
VALUES
(10011, 101, 'Savings', 55000, 'Hyderabad');
SELECT *
FROM Account;
```

Output: -



Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
10006	106	Savings	65000.00	Delhi
10007	107	Current	150000.00	Mumbai
10008	108	Savings	35000.00	Pune
10009	109	Savings	85000.00	Hyderabad
10010	110	Current	110000.00	Vijayawada
10011	101	Savings	55000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

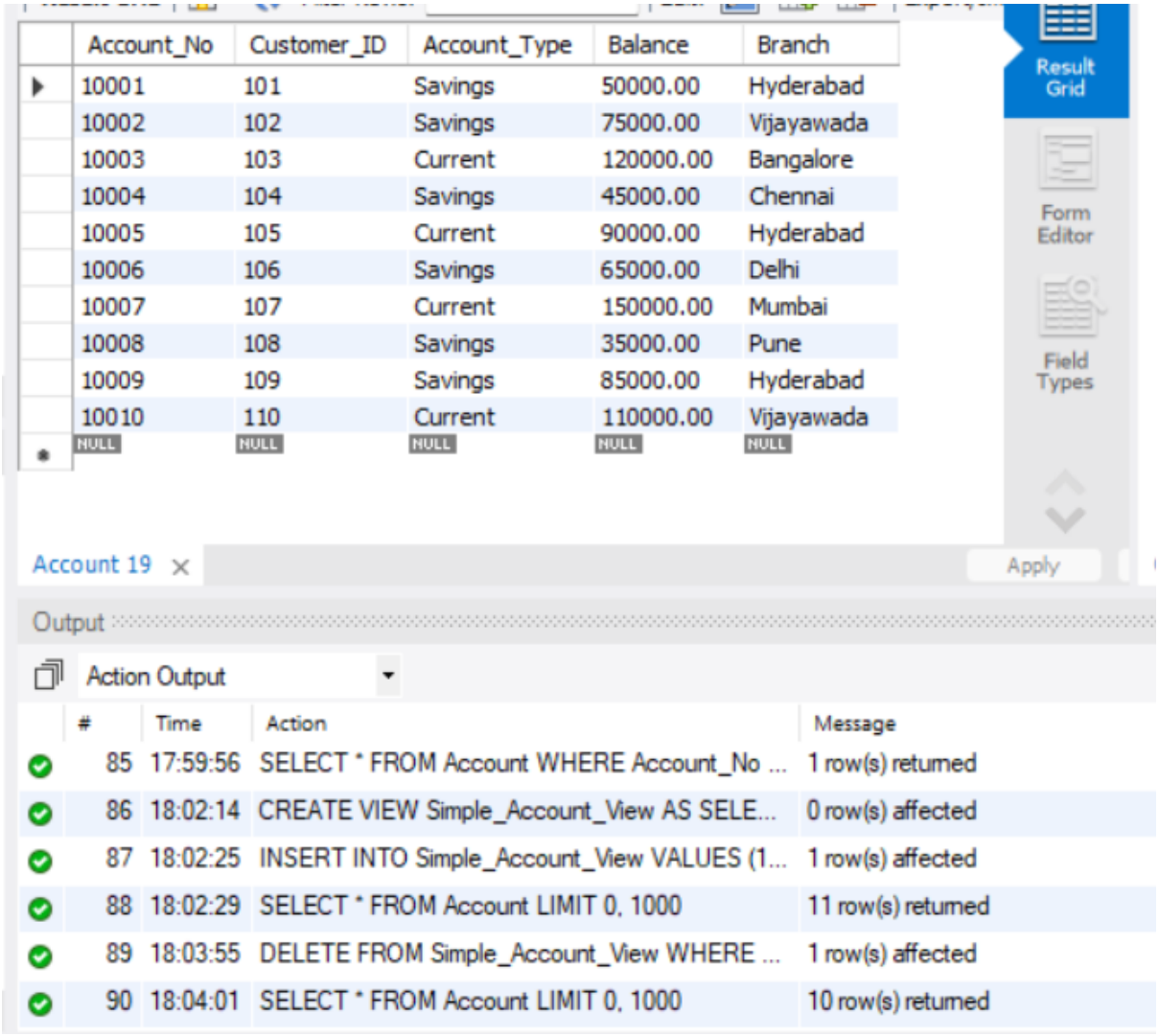


85	17:59:56	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned
86	18:02:14	CREATE VIEW Simple_Account_View AS SELE...	0 row(s) affected
87	18:02:25	INSERT INTO Simple_Account_View VALUES (1...	1 row(s) affected
88	18:02:29	SELECT * FROM Account LIMIT 0, 1000	11 row(s) returned

Query 23: -DELETE THROUGH VIEW

```
DELETE FROM Simple_Account_View
WHERE Account_No = 10011;
SELECT *
FROM Account;
```

Output: -



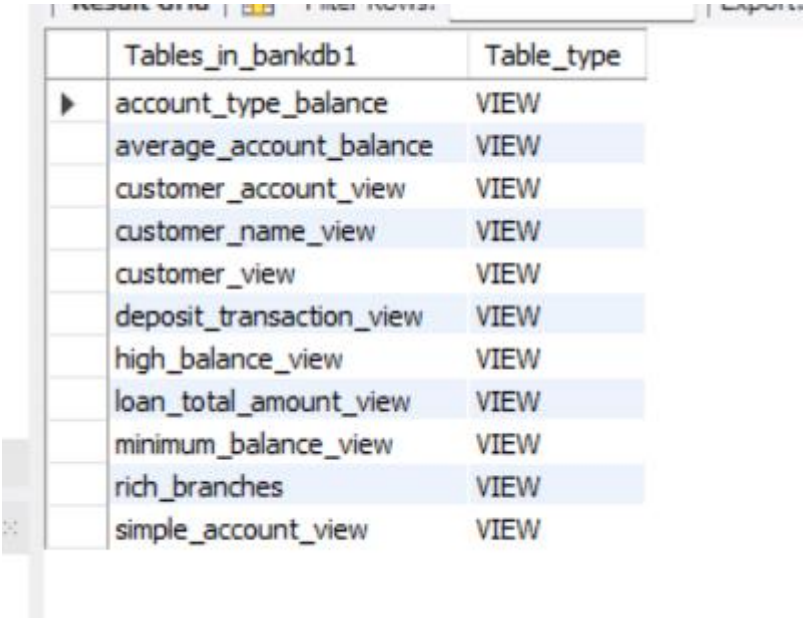
Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
10006	106	Savings	65000.00	Delhi
10007	107	Current	150000.00	Mumbai
10008	108	Savings	35000.00	Pune
10009	109	Savings	85000.00	Hyderabad
10010	110	Current	110000.00	Vijayawada
NULL	NULL	NULL	NULL	NULL

#	Time	Action	Message
85	17:59:56	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned
86	18:02:14	CREATE VIEW Simple_Account_View AS SELE...	0 row(s) affected
87	18:02:25	INSERT INTO Simple_Account_View VALUES (1...	1 row(s) affected
88	18:02:29	SELECT * FROM Account LIMIT 0, 1000	11 row(s) returned
89	18:03:55	DELETE FROM Simple_Account_View WHERE ...	1 row(s) affected
90	18:04:01	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned

Query 24: -VIEW INFORMATION

```
SHOW FULL TABLES
WHERE TABLE_TYPE = 'VIEW';
```

Output: -



Tables_in_bankdb1	Table_type
account_type_balance	VIEW
average_account_balance	VIEW
customer_account_view	VIEW
customer_name_view	VIEW
customer_view	VIEW
deposit_transaction_view	VIEW
high_balance_view	VIEW
loan_total_amount_view	VIEW
minimum_balance_view	VIEW
rich_branches	VIEW
simple_account_view	VIEW